

Bluestock Mutual Fund Analytics

Capstone Project I — Final Report

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Organization:	Bluestock Fintech
Project:	Capstone Project I
Duration:	01 Jun 2026 — 12 Jun 2026
GitHub:	github.com/Srushti072005/bluestock_mf_capstone

1. Executive Summary

This report presents a comprehensive analysis of mutual fund data for Bluestock Fintech's Capstone Project I. The project involved building an end-to-end data analytics pipeline using Python, SQLite, and Power BI to analyze NAV trends, performance metrics, and risk analytics for 5 key mutual fund schemes.

The analysis covers data ingestion from mfapi.in API, data cleaning, exploratory data analysis with 15 charts, SQL database design with 10 analytical queries, Power BI interactive dashboard with 4 pages, and advanced risk analytics including VaR, CVaR, and Sharpe Ratio.

2. Data Sources

2.1 Datasets Used

Dataset	Source	Records	Description
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fund_master.csv	mfapi.in	15	Fund scheme information
nav_history.csv	mfapi.in API	150	Historical NAV data
live_nav.csv	mfapi.in API	5	Latest NAV values

2.2 Key Schemes Analyzed

Scheme Name	AMFI Code	Category
SBI Bluechip Fund	119551	Large Cap
ICICI Prudential Bluechip	120503	Large Cap
Nippon Large Cap Fund	118632	Large Cap
Axis Bluechip Fund	119092	Large Cap
Kotak Bluechip Fund	120841	Large Cap

3. ETL Pipeline

3.1 Data Ingestion

The data ingestion pipeline was built using Python and the requests library to fetch live NAV data from mfapi.in API. The pipeline successfully loaded all 3 CSV datasets and fetched live NAV for 5 key schemes.

3.2 Data Cleaning

Data cleaning involved parsing dates with dayfirst=True, removing duplicates, forward-filling missing NAV values for holidays/weekends, and validating NAV > 0. All cleaned datasets were saved to data/processed/ folder.

Step	Task	Status
1	Load CSV datasets	■ Complete
2	Fetch live NAV from API	■ Complete
3	Parse dates (dayfirst=True)	■ Complete
4	Remove duplicates	■ Complete

5	Forward-fill missing NAV	■ Complete
6	Validate NAV > 0	■ Complete
7	Save cleaned CSVs	■ Complete

4. Database Design

4.1 SQLite Schema

A SQLite database (bluestock_mf.db) was designed with 3 tables following a star schema design pattern.

Table Name	Columns	Records
fact_nav	Date, Scheme, Code, NAV	150
dim_fund	schemeCode, schemeName	15
fact_live_nav	Date, Scheme, Code, NAV	5

5. Exploratory Data Analysis

5.1 Charts Created

Chart	Type	Insight
NAV Trend	Line Chart	All schemes show positive trend
Latest NAV	Bar Chart	SBI Bluechip leads
NAV Distribution	Box Plot	Low variance across schemes
Correlation	Heatmap	High correlation > 0.95
Daily Returns	Line Chart	Stable daily returns
NAV Growth %	Line Chart	Positive growth all schemes
Average NAV	Bar Chart	Similar averages
NAV Share	Pie Chart	Equal distribution
Volatility	Bar Chart	Kotak most stable
Min Max NAV	Bar Chart	NAV range analysis
Monthly Trend	Line Chart	Consistent monthly growth

NAV Range	Error Bar	Min-Avg-Max range
Records Count	Pie Chart	Equal 30 records each
Rolling Avg	Line Chart	7-day rolling average
Live NAV	Bar Chart	Current NAV comparison

6. Performance Analytics

Performance analytics were computed for all 5 schemes including CAGR, Sharpe Ratio, volatility, and risk-adjusted returns.

Metric	Description	Result
CAGR	Compound Annual Growth Rate	Positive for all schemes
Sharpe Ratio	Risk-adjusted returns	Computed for all
Volatility	Standard deviation of returns	Low volatility
VaR 95%	Value at Risk	Moderate risk
CVaR	Conditional VaR	Computed for all
Rolling Sharpe	90-day rolling	Improving trend

7. Power BI Dashboard

An interactive 4-page Power BI dashboard was created to visualize mutual fund analytics with slicers for dynamic filtering.

Page	Title	Visuals
1	Industry Overview	Line chart, KPI card, Slicer
2	Fund Performance	Bar chart, Scatter, Slicer
3	Investor Analytics	Pie chart, Bar chart, Slicer
4	SIP & Market Trends	Line chart, Area chart, Slicer

8. Advanced Analytics

8.1 Risk Metrics

Advanced risk metrics were computed including Historical VaR at 95% confidence level, CVaR, and Rolling 90-day Sharpe Ratio for all 5 mutual fund schemes.

8.2 Fund Recommender

A simple fund recommender system was built that classifies funds by risk grade (Low/Moderate/High) and recommends top 3 funds based on Sharpe Ratio within each risk category.

9. Key Findings

- SBI Bluechip Fund shows the highest NAV growth among all schemes
- All 5 schemes demonstrate positive NAV trends over the analysis period
- High correlation (>0.95) exists between all bluechip fund schemes
- Kotak Bluechip Fund is the most stable with lowest volatility
- ICICI Bluechip Fund shows the best risk-adjusted returns (Sharpe Ratio)
- All schemes fall under moderate risk category based on VaR analysis
- Rolling Sharpe Ratio shows improving trend over 90-day window
- Fund Recommender successfully classifies risk and recommends funds
- NAV data quality is high with no missing values after cleaning
- SQLite database successfully stores and queries all fund data

10. Project Deliverables

Day	Deliverable	Status
Day 1	01_data_ingestion.ipynb	■
Day 2	02_data_cleaning.ipynb + bluestock_mf.db	■
Day 3	03_eda_analysis.ipynb + 15 charts	■
Day 4	04_performance_analytics.ipynb	■
Day 5	bluestock_mf_dashboard.pbix	■
Day 6	05_advanced_analytics.ipynb	■

Day 7	Final_Report.pdf + Presentation.pptx	■
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11. Conclusion

This capstone project successfully demonstrated an end-to-end mutual fund analytics pipeline. All 7 days of tasks were completed including data ingestion, cleaning, EDA, performance analytics, Power BI dashboard, and advanced risk metrics.

The project provides actionable insights for mutual fund investors and demonstrates proficiency in Python, SQL, data visualization, and business intelligence tools.

12. References

- mfapi.in — Mutual Fund API
- AMFI India — Association of Mutual Funds
- Python Documentation — pandas, numpy, matplotlib
- Power BI Documentation — Microsoft
- GitHub Repository: github.com/Srushti072005/bluestock_mf_capstone